

Wellington Santos

Principal Backend & Infrastructure Engineer

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PROFESSIONAL SUMMARY

Principal Backend & Infrastructure Engineer with **8+** years across e-commerce, retail, and healthcare IT, designing and scaling production AI agents and cloud infrastructure. Deep expertise in Kubernetes orchestration, infrastructure-as-code with Terraform, and building resilient microservices on AWS and GCP. Proven track record shipping production agentic systems, establishing observability standards, and leading technical practices in client-facing consulting environments.

SKILLS

Languages: Python, Go, TypeScript, JavaScript, Bash, SQL, HTML, CSS

Backend: Node.js, FastAPI, Django, gRPC, NestJS, Express, RabbitMQ, Kafka

Generative AI: LangChain, LlamaIndex, OpenAI API, Anthropic API, LangGraph, CrewAI, Pinecone, ChromaDB, Weaviate, MLflow

Database: PostgreSQL, Redis, MongoDB, DynamoDB, Elasticsearch, Cloud Firestore

Testing: Pytest, Jest, Playwright, Cypress, k6, JMeter

Cloud & DevOps: Kubernetes, Terraform, Docker, AWS, GCP, Azure, AWS Lambda, CloudRun, Helm, GitHub Actions, Jenkins, GitLab CI/CD, Prometheus, Grafana, ELK Stack, Datadog, CloudWatch, OpenTelemetry

Others: RESTful API, GraphQL, Feature flags, Canary deployments, Blue-green deployment, Git, GitHub, GitLab, Agile, Scrum, Jira, Confluence

EXPERIENCE

Fanatics

Oct 2023–Present

Senior Backend Engineer

- Led the design and deployment of agentic infrastructure for personalized sports fan engagement, building **LangChain**-orchestrated agents that integrated with **OpenAI API** to power real-time recommendations and customer interactions across a high-volume commerce platform.
- Architected a distributed microservices backend using **Python**, **FastAPI**, and **Kubernetes**, handling 50k+ concurrent users during live sporting events with sub-second latency requirements and 99.95% uptime SLAs.
- Established observability standards across the platform by implementing **Prometheus** metrics, **Grafana** dashboards, and structured logging with the **ELK Stack**, enabling rapid incident response that reduced mean time to resolution by **42%** for backend issues.
- Designed and rolled out **feature flags** with staged canary deployments using **GitHub Actions** and **Terraform**, allowing safe testing of agent behavior changes and new sports betting features without impacting live commerce traffic.

- Collaborated with the data science team to integrate **ChromaDB** vector stores for semantic search capabilities in the personalization agent, enabling more relevant recommendations that increased average order value by **18%**.
- Refined backend API contracts across **REST**, **GraphQL**, and **gRPC** endpoints, balancing latency, payload size, and cacheability for mobile clients consuming real-time sports data and checkout flows.
- Mentored four junior backend engineers on cloud-native patterns, **Kubernetes** resource tuning, and observability best practices, establishing a reusable playbook for onboarding new microservices.
- Migrated legacy billing infrastructure from a monolith to event-driven architecture using **Kafka** and **PostgreSQL**, eliminating race conditions in payment processing that had caused **3.2%** of transactions to enter inconsistent states.
- Implemented cost-optimization initiatives across AWS cloud resources, consolidating underutilized instances and rightsizing **DynamoDB** capacity, cutting infrastructure spend by **24%** while maintaining performance.
- Led security hardening of the API gateway and backend services, implementing mutual TLS, request signing, and rate-limiting, reducing unauthorized API calls by **94%**.

Walmart

Sep 2022–Oct 2023

Senior Backend Engineer

- Spearheaded the design and implementation of a real-time inventory synchronization service using **Python**, **Kafka**, and **Redis**, connecting 5,000+ brick-and-mortar stores to the e-commerce platform and reducing inventory inconsistencies by **67%**.
- Owned the infrastructure-as-code migration from CloudFormation to **Terraform**, establishing reusable modules for **AWS** resource provisioning across development, staging, and production environments, decreasing deployment time from 90 minutes to **22 minutes**.
- Built and maintained **Kubernetes** clusters hosting the fulfillment orchestration microservices, configuring **Helm** charts for zero-downtime deployments and implementing auto-scaling policies that handled 3x peak traffic without manual intervention.
- Designed a distributed **OpenTelemetry**-instrumented observability layer across 18 backend microservices, routing traces and metrics to **Datadog**, which enabled the team to detect a cross-service memory leak within 4 hours instead of the previous 2-day turnaround.
- Collaborated with the payments team to architect a fault-tolerant transaction log service using **PostgreSQL** and event sourcing, ensuring that every payment attempt was recoverable even during partial outages, achieving **99.99%** transaction durability.
- Implemented blue-green deployment strategy for high-risk payments service, coordinating **GitHub Actions** pipelines and **feature flags** to safely roll out changes to 10% of traffic before full rollout, eliminating production payment failures for 8 consecutive months.
- Partnered with security and compliance teams to implement encryption at rest and in transit across all backend services, passing a SOC 2 Type II audit and enabling Walmart to serve regulated financial services customers.

- Mentored a team of 5 backend engineers on backend architectural patterns, teaching trade-offs between **REST** and **GraphQL** APIs for different client types, and establishing code review standards that improved test coverage from 61% to 89%.

E-POT

Jun 2018–July 2022

Full Stack Engineer

- Built medical data integration pipelines for healthcare IT clients, implementing **Python**, **FastAPI**, and **PostgreSQL** backend services that ingested and normalized HL7/FHIR records from 50+ hospital systems, making previously siloed patient records searchable across enterprise systems.
- Designed and deployed containerized microservices using **Docker** and **Kubernetes**, establishing the company's first standardized deployment platform that reduced new service onboarding time from 2 weeks to 3 days for SE staffing projects.
- Architected **Terraform** infrastructure templates for common deployment patterns—web apps, databases, message queues—enabling the SE team to provision client environments without waiting for IT, reducing engagement setup delays by 5 days per project.
- Implemented structured logging and **Prometheus** metrics across the manufacturing IT division's operational dashboards, surfacing equipment health anomalies that previously went unnoticed, improving mean time to repair by **31%** for factory floor systems.
- Developed a custom feature-flag service using **Redis** and **Node.js** to support A/B testing and staged rollouts for banking clients undergoing core system migrations, allowing safe validation of new transaction processing logic.
- Collaborated with healthcare compliance teams to design secure **REST API** gateways with encryption, audit logging, and role-based access control, enabling the company to serve HIPAA-regulated clients and expand into fintech systems.
- Led a proof-of-concept AI solution prototype using **LangChain** and **OpenAI API** to automate medical coding and billing record review, demonstrating 87% accuracy and laying the foundation for the company's AI Solutions practice.
- Mentored 3 junior full-stack engineers transitioning from staffing roles to product development, teaching cloud-native patterns and infrastructure best practices that elevated the team's technical credibility with clients.

EDUCATION

University of Tokyo

Apr 2014–Mar 2018

Bachelor of Science in Computer Science